





$\text{귀납법} \Rightarrow \sum |a_0\rangle_n \dots |a_{i-1}\rangle_n U_m (NOT_n (|a_i\rangle_n |0\rangle_n) \dots = |a_0\rangle_n \dots |a_{i-1}\rangle_n U_m (|a_i\rangle_n |a_i\rangle_n) = |a_0\rangle_n \dots |a_{i-1}\rangle_n$   
 $|a_i\rangle_n |a_i \bmod N\rangle_n |0\rangle_n \dots = |a_0\rangle_n \dots |a_{i+1}\rangle_n |0\rangle_n \dots \quad \text{이므로} \quad \text{성립} \quad \text{loop } (U_m: O(n^2)) \therefore O(n^3)$

$(-U_a: |\psi\rangle = |c\rangle_n |a\rangle_n |0\rangle_n^{\otimes n-1} |1\rangle_n |t\rangle_n, U_p \sum |\psi\rangle = |c\rangle_n |a_0\rangle_n \dots |a_{n-1}\rangle_n |1\rangle_n |t\rangle_n, i = 1 \sim n$

$|\psi\rangle = \text{SWAP}_{i-1} |\psi\rangle, |\psi_{n+1}\rangle |\psi_{n+2}\rangle = U_m |\psi_{n+1}\rangle |\psi_{n+2}\rangle, |\psi\rangle = \text{SWAP}_{i-1} |\psi\rangle$

$P \in) (\text{SWAP}_i |\psi\rangle = |c\rangle_n \dots |a_{n-1}\rangle_n |1\rangle_n |t\rangle_n \text{ if } c_0 = 0, |c\rangle_n \dots |a_{n-1}\rangle_n |1\rangle_n |a_{n-1}\rangle_n |t\rangle_n \text{ if } c_0 = 1$

$\therefore U_m |1\rangle_n |t\rangle_n = |1\rangle_n |t\rangle_n = |1\rangle_n |a_{n-1}^{c_0} \bmod N\rangle_n = |1\rangle_n |t_1\rangle_n, U_m |a_{n-1}\rangle_n |t\rangle_n = |a_{n-1}\rangle_n |t\rangle_n \text{ or}$

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$|a_{n-1}\rangle_n |a_{n-1}^{c_0} \bmod N\rangle_n = |a_{n-1}\rangle_n |t_1\rangle_n, \text{즉} (\text{SWAP}_i: \psi = |c\rangle_n |a_0\rangle_n \dots |a_{n-1}\rangle_n |1\rangle_n |t_1\rangle_n \text{ 이다.}$

$\text{귀납법} \Rightarrow i = n: |\psi\rangle = |c\rangle_n |a_0\rangle_n \dots |a_{n-1}\rangle_n |1\rangle_n |t_n\rangle_n \therefore |c\rangle_n |t_n\rangle_n = (-U_a |c\rangle_n |t\rangle_n$

$\text{Oscilla bit: } O(n^2), U_p: O(n^2), \text{loop } (\text{SWAP}_i: O(1), U_m: O(n^2)) \therefore O(n^3) \text{ 이다.}$



$$f: \mathbb{N}^2 \rightarrow \mathbb{N} \text{ with } \bar{w}_f = 4 \quad (6.4.6) \quad n = O(\log N) \therefore O(n^3) = O((\log N)^3)$$

integer factorization problem: classical, fully analyzed Monte Carlo algorithm

$$: e^{(1+o(1))(\log N \cdot (\log \log N)^{\frac{1}{2}})}$$

$$\text{heuristic Monte Carlo algorithm: } e^{(\frac{4}{3} + o(1)) \cdot (\log N)^{\frac{1}{3}} \cdot (\log(\log N))^{\frac{2}{3}}}$$

$a \in \mathbb{Z}_n$ : select,  $d = \gcd(a, N)$  compute  $d=1 \Rightarrow$  Find order  $(N, a, 2\lceil \log N \rceil + 1) \subseteq \mathbb{Z}$

$$a^r \equiv 1 \pmod{N} \text{ if } r \text{ is a multiple of } \frac{N-1}{2} \Rightarrow \text{if } r \text{ is a multiple of } \frac{N-1}{2} \text{ then } (a^{r/2-1} / a^{r/2+1}) \equiv 0 \pmod{N} \wedge a^{r/2+1} \Rightarrow \gcd(a^{r/2+1}, N):$$

$: N$  is proper divisor with probability at least  $0.199, O((\log N)^3)$

$p$ : odd prime number,  $e \in \mathbb{N}$ ,  $p = 2^d m + 1$ ,  $2 \nmid m$ ,  $a \in \mathbb{Z}_p^*$ ,  $a^r \equiv 1 \pmod{p^e}$ ,  $2^d | r$   $\frac{r}{2^d} \equiv \frac{1}{2} \pmod{m}$

$p \nmid 1$   $\varphi(p^e) = (p-1)p^{e-1}$ ,  $g \in \mathbb{Z}_p^*$ ,  $g^{\varphi(p^e)} \equiv 1 \pmod{p^e}$   $\mathbb{Z}_p^* \rightarrow \mathbb{Z}_p^*$ ,  $\kappa \mapsto g^\kappa \pmod{p^e}$ : bijective,

$a = g^\kappa \pmod{p^e} \Rightarrow a$  is order  $r = \frac{\varphi}{\gcd(\kappa, \varphi)}$   $2 \nmid \varphi$   $\kappa$ : even  $\frac{1}{2}$ , odd  $\frac{1}{2} \therefore 2^d | r : 1/2$



$$N = \prod_{i=1}^n p_i^{e_i}, a \in \mathbb{Z}_N^*, a \bmod N \text{ of order } r: 2(r, a^{r/2} \pm 1 \bmod N) \geq 1 - 2^{1-m}$$

$$P \in \{ a_i \in \mathbb{Z}_{p_i^{e_i}}, a \in \mathbb{Z}_N^*, a \equiv a_i \bmod p_i^{e_i}, r: \text{order of } a \bmod N, r_i: \text{order of } a_i \bmod p_i^{e_i}$$

$$r = \text{lcm}(r_1, \dots, r_n), r = 2^f m, r_i = 2^{f_i} m_i, (m, m_i: \text{odd}), r: \text{odd or } r: \text{even}, a^{r/2} \equiv -1 \bmod N \Rightarrow 1 \leq i \leq m, f = f_i$$

$$r: \text{odd}: 2 \nmid r, 2(r, a^{r/2} \equiv -1 \bmod N) \Rightarrow a^f \equiv 1 \bmod p_i^{e_i}, a^{r/2} \equiv -1 \bmod p_i^{e_i} \therefore r, r_i \nmid r/2: \therefore f = f_i$$

$$\therefore r: \text{odd or } 2 \mid r, a^{r/2} \equiv -1 \bmod N \leq \forall i, 1 \leq i \leq m, f_i = f_j = \frac{1}{2^{m-1}} \therefore$$

$$1 - \frac{1}{2^{m-1}} \geq \frac{1}{2} \therefore 0.999 \cdot \frac{1}{2} \approx 0.199, O(\log N |^3)$$

$$2 \nmid N, a, b \in \mathbb{Z}_N^*, r: \text{order of } a \bmod N, a^t \equiv b \bmod N: t: \text{discrete logarithm of } b \text{ to base } a \bmod N, \text{ we may assume } r \text{ is prime by the Pohlig-Hellman algorithm, } t > 1$$

$$n \in \mathbb{N}, U_a, U_b: |x\rangle_n \rightarrow \begin{matrix} |u \text{ or } b\rangle_n \bmod N \\ |x\rangle_n \end{matrix} \begin{matrix} 0 \leq x < N \\ N \leq x < 2N \end{matrix}, U_a \text{ is eigenvalue: } a^{2\pi i \cdot \frac{t}{r}}, U_b \text{ is eigenvalue: } a^{2\pi i \cdot \frac{t}{r}}$$

$$0 \leq k < r, \text{ by quantum phase estimation, } (x, y) \in \mathbb{Z}_m \times \mathbb{Z}_r, K = \lfloor \frac{r}{2^n} \rfloor, K \neq \bmod r = \lfloor \frac{r}{2^n} \rfloor: t = K^2 \lfloor \frac{r}{2^n} \rfloor \bmod r$$



$n = \lceil \log_2 r \rceil + 1$  : probability  $\frac{64(r-1)}{r^4}$ ,  $O((\log(V))^3)$

$P(t) |\psi_0\rangle = |0\rangle_n |0\rangle_n |1\rangle_n$ ,  $|\psi_1\rangle = \left(\frac{107+117}{\sqrt{2}}\right)^{\otimes n} \left(\frac{107+117}{\sqrt{2}}\right)^{\otimes n} |1\rangle_n$ ,  $(-U_a \text{ operator } |\psi_{2a}\rangle = \frac{1}{\sqrt{r}} \sum_{k=0}^{r-1} (|\psi_k(\frac{a}{r})\rangle$   
 $\cdot (\sum_{k=0}^{r-1} |\psi_k\rangle) \cdot |u_n\rangle)$   $(-U_b \text{ operator: } |\psi_{2b}\rangle = \frac{1}{r} \sum_{k=0}^{r-1} (|\psi_k(\frac{a}{r})\rangle |\psi_k(\frac{b}{r})\rangle |u_k\rangle)$ ,  $|t_3\rangle = (\frac{1}{r}, |\psi_k(\frac{a}{r})\rangle, |\psi_k(\frac{b}{r})\rangle)_{0 \leq k < r}$

apply  $QFT_n^{-1}$  :  $|\psi_4\rangle = (\frac{1}{r}, QFT_n^{-1} |\psi_k(\frac{a}{r})\rangle QFT_n^{-1} |\psi_k(\frac{b}{r})\rangle)_{0 \leq k < r}$  probability  $\frac{64}{r^4}$ ,  $(x,y) \in \mathbb{Z}_m \times \mathbb{Z}_n$ ,  
 $|\Delta(\frac{a}{r}, n)\rangle | \leq \frac{1}{2n} < \frac{1}{2r}$ ,  $|\Delta(\frac{kt_{mod}r}{r}, n, y)\rangle | \leq \frac{1}{2n} < \frac{1}{2r}$   $0 \leq z$   $K = \lfloor \frac{r}{2n} \rfloor$ ,  $kt_{mod}r = \lfloor \frac{r}{2n} \rfloor$

Classical public-key cryptography is vulnerable - Quantum-resistant cryptography

Group  $G$ ,  $H < G$ , set  $X$ ,  $f: G \rightarrow X$ ,  $\forall g, g' \in G$ ,  $gH = g'H \Leftrightarrow f(g) = f(g')$   $\iff f$  hides subgroup  $H$

$\therefore f$  has the same value for all elements of a coset of  $H$ ,

input:  $f$ , output: finite generating system for  $H$

Deutsch:  $f: \{0,1\} \rightarrow \{0,1\}$ ,  $f$  is constant:  $f$  hides  $\{0,1\}$ ,  $f$  is balanced:  $f$  hides  $\{0\}$



Simon's problem:  $f: \{0,1\}^n \rightarrow \{0,1\}^n$ ,  $\vec{z} \in \{0,1\}^n$ ,  $\vec{z} \neq \vec{0}$ ,  $\forall \vec{x}, \vec{y} \in \{0,1\}^n$ ,  $\vec{y} = \vec{x} \oplus \vec{z} \Leftrightarrow f(\vec{x}) = f(\vec{y})$   
 $H = \{\vec{0}, \vec{z}\}$ ; Generating system  $(\vec{z})$ ,  $(\vec{0}, \vec{z})$

$N \geq 3$ ,  $2 \nmid N$ ,  $a \in \mathbb{Z}_N$ ,  $\gcd(a, N) = 1$ .  $f: (\mathbb{Z}_N, +) \rightarrow \mathbb{Z}_N$ ,  $f(i) = a^i \pmod{N}$ ,  $H = r\mathbb{Z} : (r)$

$a, b \in \mathbb{Z}_N^*$ ,  $r$ : order of  $a \pmod{N}$ .  $f: (\mathbb{Z}_r \times \mathbb{Z}_r, +) \rightarrow \mathbb{Z}_N$ ,  $(x, y) \mapsto a^x b^y$ ,  $H = \mathbb{Z}_r(1, -t)$

$r, t, m$  s.t.  $H = \mathbb{Z}_r(1, -t) \subseteq$  generating system  $((x_0, y_0), \dots, (x_{m-1}, y_{m-1}))$ ,  $d = \gcd(x_0, \dots, x_{m-1})$

$\gcd(d, r) = 1$ .  $t = d^{-1} \gcd(y_0, \dots, y_{m-1}) \pmod{r}$

$P \in \mathbb{Z} \exists u_0, \dots, u_{m-1} \in \mathbb{Z}$ ,  $\sum_{i=0}^{m-1} u_i (x_i, y_i) \equiv (1, -t) \pmod{r}$ ,  $(x_i, y_i) \in \mathbb{Z}_r(1, -t)$ ,  $\sum_{i=0}^{m-1} u_i x_i (1, -t) \equiv (1, -t) \pmod{r}$

$\therefore x'_i = x_i / d \pmod{r}$ ,  $d \sum_{i=0}^{m-1} u_i x'_i \equiv 1 \pmod{r} \because \gcd(d, r) = 1$ ,  $\gcd(y_0, \dots, y_{m-1}) \equiv td \pmod{r}$